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Hospital Pakar Universiti Sains Malaysia (HPUSM)

HOSPITAL PAKAR UNIVERSITI SAINS MALAYSIA
Hospital Pengajar Universiti
Kementerian Pendidikan Tinggi

GUIDE TO ANTIMICROBIAL THERAPY

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ANTIMICROBIAL STEWARDSHIP
PROGRAMME
(ASP)

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LIST OF ABBREVIATIONS

AKI	Acute kidney injury	MAP	Mean Arterial Pressure
AmpC	AmpC Beta-lactamase	MBLs	Metallo-Beta-lactamases
CAUTI	Catheter-associated urinary tract infections	MDR	Multidrug-resistant
CDI	Clostridium difficile infections	MDR PAE	Multidrug-resistant <i>Pseudomonas aeruginosa</i>
CLABSI	Central line-associated bloodstream infections	MDR AB	Multidrug-resistant <i>Acinetobacter baumannii</i>
COPD	Chronic Obstructive pulmonary Disease	MRCONS	Methicillin-resistant coagulase-negative <i>Staphylococci</i>
CR	Carbapenem-resistant	MRSA	Methicillin-resistant <i>Staphylococcus aureus</i>
CR PAE	Carbapenem-resistant <i>Pseudomonas aeruginosa</i>	MSSA	Methicillin-sensitive <i>Staphylococcus aureus</i>
CRAB	Carbapenem resistant <i>Acinetobacter baumannii</i>	NDM	New Delhi metallo-Beta-lactamases
CRE	Carbapenem-resistant Enterobacterales	NMTT	N-methylthiotetrazole
CRI	Catheter-related infection	OXA	Oxacillinase
DAIR	Debridement, antibiotics, and implant retention	PBP2a	Penicillin-binding protein 2a
DTR	Difficult-to-treat resistance	SIRS	Systemic inflammatory response syndrome
ESBL	Extended-spectrum Beta-lactamase	SSI	Surgical Site Infection
ESBL-E	Extended-spectrum Beta-lactamase-producing Enterobacterales	STD	Sexual Transmitted Disease
FBC	Full blood count	UTI	Urinary tract infection
GPC	Gram Positive Cocci	VAP	Ventilator-associated pneumonia
HAI	Hospital-acquired infections	VIM	Verona integron-encoded metallo-Beta-lactamase
HAP	Hospital-acquired pneumonia	VISA	Vancomycin-intermediate <i>Staphylococcus aureus</i>
HCAI	Healthcare-associated Infection	VRE	Vancomycin-resistant <i>Enterococci</i>
HPUSM	Hospital Pakar Universiti Sains Malaysia	VRSA	Vancomycin-resistant <i>Staphylococcus aureus</i>
ICU	Intensive care unit	VS	Versus
IMPs	Imipenem-hydrolyzing metallo-Beta-lactamases	XDR	Extensively drug-resistant
INR	International Normalized Ratio	XDR AB	Extensively drug-resistant <i>Acinetobacter baumannii</i>
IV	Intravenous	XDRO	Extensively drug-resistant organism
KPCs	<i>Klebsiella pneumoniae</i> Carbapenemases		

THE ASP GUIDE TO ANTIMICROBIAL THERAPY:

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CHAPTER 1

INFECTION, SEPSIS AND SEPTIC SHOCK

Infection is said to have occurred when there is an inflammation as a result of invading microorganism. This inflammatory response could be localised or systemic. Sepsis is a systemic or whole-body inflammatory response to an infection. Specifically, sepsis can be defined as a potentially life-threatening organ dysfunction caused by a dysregulated host response to infection.

Fever can be a cardinal feature of sepsis. Other common manifestations of sepsis include tachycardia, tachypnoea, and confusion. Organ or source specific manifestations should always be sought, for example cough for pneumonia, or dysuria for a UTI. In a very young, old, and people with a weakened immune system, there may be no symptoms of a specific infection and the body temperature may be low or normal rather than high.

In short, **sepsis** is a condition when there is an infection (eg, infected wound) with systemic manifestation of inflammations (fever, chills & rigors, severe lethargy, temperature of more than 38°C, tachycardia and/or leucocytosis).

At the other end of the spectrum, **septic shock** occurs when there is sepsis AND tissue hypoperfusion (eg, cold clammy peripherals, crt > 2s, hypotension and/ or lactate increasing).

Septic shock is a medical emergency and thus requires immediate and accurate interventions. These interventions are grouped together in sepsis bundle to guide clinicians. The five components of sepsis bundle are:

- Measure lactate level. Remeasure if initial lactate is > 2mmol/L.
- Obtain blood cultures prior to the administration of the antibiotics.
- Administer broad-spectrum antibiotic(s).
- Begin administration of 30ml/kg crystalloid for hypotension OR lactate > 4mmol/L.
- Apply vasopressor if the patient remain hypotensive during OR after fluid resuscitations to maintain MAP > 65mmHG.

Table 1 Bundle elements with strength of recommendations and under-pinning quality of evidence [12, 13]

Bundle element	Grade of recommendation and level of evidence
Measure lactate level. Re-measure if initial lactate is > 2 mmol/L	Weak recommendation, low quality of evidence
Obtain blood cultures prior to administration of antibiotics	Best practice statement
Administer broad-spectrum antibiotics	Strong recommendation, moderate quality of evidence
Rapidly administer 30 ml/kg crystalloid for hypotension or lactate $\geq$ 4 mmol/L	Strong recommendation, low quality of evidence
Apply vasopressors if patient is hypotensive during or after fluid resuscitation to maintain MAP $\geq$ 65 mm Hg	Strong recommendation, moderate quality of evidence

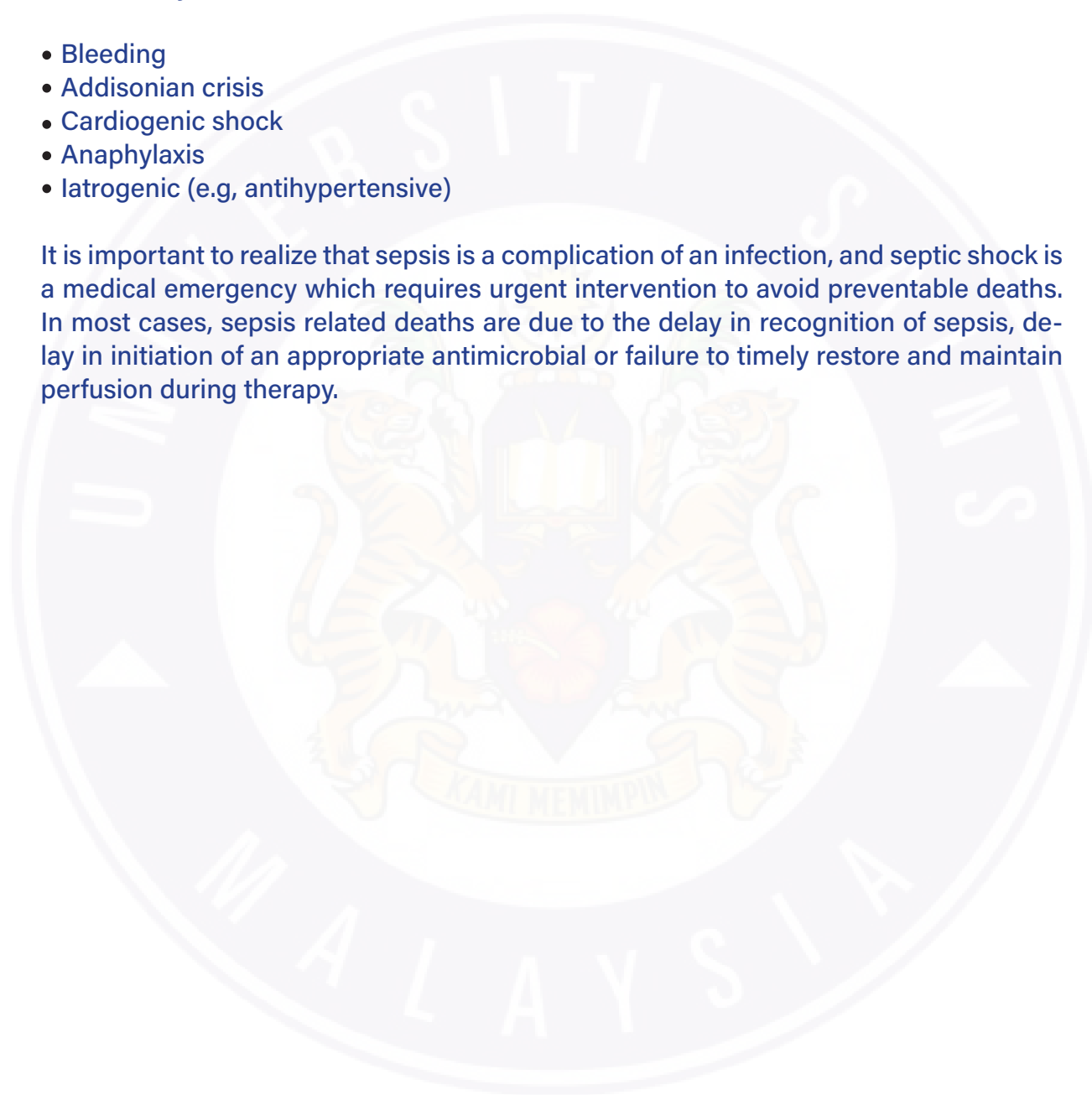
Adapted from: Intensive Care Medicine (2018) 44:925-928

Initiation of fluid resuscitation **AND** antibiotic therapy must be completed within 1-hour. **ALL** five components of the bundle **MUST** be completed within 3-hours.

The patient must have received at least 60ml/kg of fluid at 6-hours to be considered as receiving enough fluid resuscitation. Hence, the patients who have received adequate fluid resuscitation and yet remained hypotensive need to be investigated and or treated concurrently for other POSSIBLE causes of shock. Other causes of shock are;

- Bleeding
- Addisonian crisis
- Cardiogenic shock
- Anaphylaxis
- Iatrogenic (e.g, antihypertensive)

It is important to realize that sepsis is a complication of an infection, and septic shock is a medical emergency which requires urgent intervention to avoid preventable deaths. In most cases, sepsis related deaths are due to the delay in recognition of sepsis, delay in initiation of an appropriate antimicrobial or failure to timely restore and maintain perfusion during therapy.



CHAPTER 2

A GUIDE TO ANTIMICROBIAL THERAPY FOR COMMON & IMPORTANT INFECTIONS

Location/ Severity	Common Pathogens	First choice/ Preferred	Alternative	Notes
HEAD - CENTRAL NERVOUS SYSTEM				
Brain abscess	<i>Streptococcus</i> sp. <i>Bacteroides</i> . <i>Staphylococcus aureus</i> <i>Enterobacteriaceae</i>	IV Ceftriaxone 2g q12h PLUS IV Metronidazole 500mg q8h	IV Cefotaxime 2g q4-6h PLUS IV Metronidazole 500mg q8h ^{1,2}	DURATION: 4 to 8 WEEKS with minimum 2 weeks IV therapy ²
Meningitis	<i>N. meningitidis</i> <i>H. influenzae</i> <i>Strep. pneumoniae</i> <i>L. monocytogenes</i>	IV Ceftriaxone 2g q12h PLUS IV Ampicillin 2g q4h ^{2,3,4}		Add IV Dexamethasone 0.15mg/kg q6h for 2 to 4 days ¹ DURATION: 7 to 21 days ⁷
NECK - EAR, NOSE AND THROAT				
Acute pharyngitis/ tonsillitis	Usually viral <i>Strep. pyogenes</i> Other strep. sp	Oral Phenoxyethyl penicillin (Penicillin V) 500mg q6h or 1000mg q12h ^{1,2}	Oral Erythromycin 800mg q12h ²	DURATION: 5 to 10 days ^{1,2}
Otitis media	<i>H. influenzae</i> <i>Strep. pneumoniae</i>	Oral Amoxicillin 500-1000mg q8h ^{1,2}	Oral Amoxicillin-Clavulanic Acid 625mg q8h ²	DURATION: 5 to 10 days ^{1,2,3}
Sinusitis	<i>H. influenzae</i> <i>Strep. pneumoniae</i> <i>Staphylococcus aureus</i>			DURATION: 5 to 7 days ^{1,2,3}
LUNGS - RESPIRATORY SYSTEM				
Acute Exacerbation of COPD	<i>H. influenzae</i> , <i>M. catarrhalis</i> , <i>S. pneumoniae</i>	IV Amoxicillin-Clavulanic Acid 1.2g q8h PLUS Oral Azithromycin 500mg q24h ²		DURATION: 5 to 7 days ²

Pneumonia (Community-acquired) Mild to moderate severity CURB65 = 0 to 1	<i>S. pneumoniae</i> <i>H. influenzae</i> <i>M. catarrhalis</i>	Oral Amoxicillin-Clavulanic acid 625mg q8h		CURB 65: Confusion Urea RespRate BP Age > 65 DURATION: 5 to 7 days ²
Pneumonia (Community-acquired) Moderate severity CURB65 = 2 to 3	<i>S. pneumoniae</i> <i>Mycoplasma</i> <i>H. influenzae</i> <i>Staphylococcus aureus</i> <i>Legionella pneumophila</i> <i>Chlamydia pneumoniae</i>	IV Amoxicillin-Clavulanic acid 1.2g q8h PLUS Oral Azithromycin 500mg q24h ²	IV Ceftriaxone 2g q24h PLUS Oral Azithromycin 500mg q24h ²	
SEVERE pneumonia (CURB65 > 3)	<i>S. pneumoniae</i> <i>H. influenzae</i> <i>Staphylococcus aureus</i> <i>Legionella pneumophila</i> <i>Enterobacteriaceae</i> <i>Pseudomonas</i>		IV Piperacillin-tazobactam 4.5g q6-8h* PLUS Oral/IV Azithromycin 500mg q24h ² *Risk of <i>Pseudomonas</i> infection	
Pneumonia (Aspiration)	<i>S. pneumoniae</i> <i>Staphylococcus aureus</i> <i>Anaerobes</i> <i>K.pneumoniae</i>	IV Amoxicillin-Clavulanic acid 1.2g q8h	IV Ceftriaxone 2g q24h PLUS/- IV Metronidazole 500mg q8h ^{2,4}	DURATION: 5 to 7 days ^{2,4}
Lung abscess	<i>Anaerobes</i> <i>S. pneumoniae</i> <i>Strep. milleri</i> <i>Staphylococcus aureus</i> <i>K.pneumoniae</i> <i>Nocardia</i>	IV Amoxicillin-Clavulanic acid 1.2g q6-8h ^{2,4}	IV Ceftriaxone 2g q24h PLUS IV Metronidazole 500mg q8h ^{2,4}	DURATION: 28 to 42 days (After afebrile 4-5 days, change to oral therapy)
Lung empyema	<i>S. pneumoniae</i> <i>H. influenzae</i> <i>Anaerobic strep.</i> <i>Strep. milleri</i> <i>Bacteroides sp.</i> <i>Staphylococcus aureus</i> <i>Enterobacteriaceae</i> <i>M. tuberculosis</i>			Require drainage of pus via chest tube

HEART - CARDIOVASCULAR SYSTEM

Infective endocarditis² (Empirical Therapy for native valve)	<i>Strep. viridans</i> Other <i>Strep.</i> <i>Staphylococcus sp.</i> <i>Enterococcus</i>	IV Ampicillin 2g q4h PLUS IV Gentamicin 3mg/kg q24h PLUS/MINUS IV Cloxacillin 2g q4h ²	(Penicillin allergy) IV Vancomycin 15-20mg/kg q8-12h PLUS IV Gentamicin 3mg/kg q24h ^{1,2}	At least 2 sets of blood cultures 10mls each is MANDATORY prior to initiation of antibiotics.
Prosthetic valve Infective endocarditis¹ (Empirical Therapy - culture pending)	<i>Staph. Epidermidis</i> <i>Staph. Aureus</i> <i>Strep. viridans</i> Other <i>Strep.</i> <i>Enterococci</i>	IV Vancomycin 15-20mg/kg q8-12h (Maximum:2g/dose) PLUS IV Gentamicin 3mg/kg q24h ^{1,2} PLUS Oral Rifampicin 300-450mg q12h ^{2,4}		DURATION: 28 to 42 days (<i>Viridan strep.</i> 28 days, <i>Enterococci sp.</i> 42 days, <i>S. aureus</i> 28 days)

ABDOMEN - GASTROINTESTINAL SYSTEM

Liver abscess	<i>Klebsiella sp.</i> , <i>Enterobacteriaceae</i> , <i>Bacteroides</i> <i>Enterococci</i> <i>Entamoeba histolytica</i> , <i>Yersinia E.</i> ,	IV Amoxicillin-Clavulanic acid 1.2g q6-8h ²	IV Ceftriaxone 2g q24h PLUS/- IV Metronidazole 500mg q8h ^{1,2}	DURATION: 14 to 42 days ²
Acute cholecystitis	<i>Enterobacteriaceae</i> <i>Enterococci</i> <i>Bacteroides</i> <i>Clostridium sp.</i>	IV Amoxicillin-Clavulanic acid 1.2g q6h ²		DURATION: 4 to 7 days ²
Ascending cholangitis				
Bacterial peritonitis		IV Amoxicillin-Clavulanic acid 1.2g q6h ²	IV Cefotaxime 2g q8h ^{1,2,4}	DURATION: 5 to 7 days ^{1,2}
<i>H.pylori</i> gastritis	<i>Helicobacter pylori</i>	Oral Amoxicillin 1g q12h PLUS Oral Clarithromycin 500 mg q12h ²	Oral Clarithromycin 500 mg q12h PLUS Oral Metronidazole 400mg q8h ^{1,3}	In combination with Oral Pantoprazole DURATION: 14 days ²
Salmonella enteritis	<i>Salmonella Typhi</i>	IV Ceftriaxone 2g q12-24h ²	Oral Ciprofloxacin 500-750mg q12h ² OR Oral Azithromycin 1g q24h ^{1,2,3}	DURATION: 7 to 14 days ^{2,3}

BLADDER - GENITOURINARY SYSTEM

(Acute cystitis) UTI - uncomplicated	<i>E. coli</i> <i>K. pneumoniae</i> <i>Proteus mirabilis</i>	Oral Nitrofurantoin 50-100mg q6h ²	Oral Cephalexin 500mg q12h ¹	DURATION: 3 to 5 days ^{1,2}
UTI - complicated E.g: Male, stones		IV Amoxicillin-Clavulanic acid 1.2g q6h ²	IV Cefuroxime 750-1500mg q8h ²	DURATION: 10 to 14 days ²
Acute Pyelonephritis	<i>E. coli</i> <i>K. pneumoniae</i> <i>Proteus mirabilis</i>	IV Amoxicillin-Clavulanic acid 1.2g q6h PLUS IV Gentamicin 5mg/kg/24h ²	IV Cefuroxime 750-1500mg q8h ²	DURATION: 7 days ²

Urosepsis	<i>E. coli</i> <i>K. pneumoniae</i> <i>Proteus mirabilis</i> <i>P. aeruginosa</i>	IV Piperacillin-Tazobactam 4.5g q8h ²	IV Cefepime 1g q8h ²	DURATION: 10 to 14 days
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FOOT - DIABETIC FOOT ULCER INFECTION

Mild		Oral Cephalexin 500mg q6h ^{1,2}	Oral Amoxicillin-Clavulanic acid 625mg q8h ²	DURATION: 7 to 14 days ²
Moderate to severe²		IV Ampicillin-Sulbactam 3g q6-8h ^{1,2} IV Ceftazidime 2g q6h ^{1,2} (if suspected melioidosis)		

SKIN AND SOFT TISSUES INFECTION

Cellulitis -mild	<i>Staph aureus</i> <i>Strep sp.</i>	Oral Cloxacillin 500mg q6h ²	Oral Amoxicillin-Clavulanic acid 625mg q8h ²	DURATION: 5 to 10 days ²
Cellulitis -moderate	<i>Staph aureus</i> <i>Strep sp.</i>	Oral Cloxacillin 500mg q6h ²	IV Ampicillin-Sulbactam 3g q6-8h ²	DURATION: 7 days ²
Cellulitis -severe	<i>Staph aureus</i> <i>Strep sp.</i>	IV Piperacillin-Tazobactam 4.5g q6-8h	IV Ampicillin-Sulbactam 3g q6-8h ²	DURATION: 7 days ²
Necrotising Fasciitis	<i>Strep sp</i> <i>Clostridia sp</i> <i>Anaerobic</i> <i>Staph aureus</i>	IV Piperacillin-Tazobactam 4.5g q6-8h ^{1,2} PLUS/MINUS IV Clindamycin 600-900mg q8h ²	IV Ampicillin-Sulbactam 3g q6h ^{4,5}	DURATION: 7 to 14 days ²
Animal/human bites		Oral Amoxicillin-Clavulanic acid 625mg q8h ^{1,2,4,5}		DURATION: 5 days ²

BONE

Acute osteomyelitis	<i>Staph aureus</i> (80%) <i>Gp A Strep</i>	IV Cloxacillin 2g q6h ²	IV Clindamycin 600mg q8h ^{1,3,5}	DURATION: 2 to 4 weeks with IV therapy followed by oral therapy. Total 6 weeks of antibiotics ²
Chronic Osteomyelitis	<i>Commonest Staph aureus</i>	Empirical treatment is not indicated. Need surgical de- bridement. Treatment based on tissue C&S.		DURATION: 6 weeks to 3 months ^{1,2}

JOINTS

Acute monoarticular -No-STD risk	<i>Staph. aureus</i> <i>Strep sp.</i>	IV Cloxacillin 2g q4-6h ²	IV Clindamycin 600mg q6h ²	May need drainage, debridement or washout. DURATION: 2 to 4 weeks with IV therapy followed by oral. Total 6 weeks of antibiotics ²
Acute monoarticular -STD risk	<i>N. gonorrhea</i> , <i>Strep sp.</i> / <i>Staph sp.</i>	IV Ceftriaxone 1g q24h ¹	IV Cefotaxime 2g q8h ¹	

SYSTEMIC INFECTIONS (UNDIFFERENTIATED)

Leptospirosis-mild		Oral Doxycycline 100mg q12h x 5-7 days ²	Oral Azithromycin 500mg q24h x 3 days ²	
Leptospirosis-moderate to severe Severe leptospirosis: Altered sensorium, AKI (Oliguria, hyperkalemia, creat 265 umol/l), respiratory insufficiency (dyspnea, pulmonary infiltrates), hypotension, arrhythmias. ⁶		IV Benzylpenicillin 1.5MU q6h ^{1,2}	IV Ceftriaxone 2g q24h ^{1,2}	DURATION: 7 to 10 days ² Add Steroid for ALL severe Leptospirosis. (Hydrocortisone 100-200mg q8h OR Methyprednisolone 0.5-1g q24h)
Rickettsial-mild		Oral Doxycycline 100mg q12h for 5 days	Oral Azithromycin 500mg q24h for 3 days	
Rickettsial-moderate to severe		IV Azithromycin 500mg q12h on Day 1 then 500mg q24h PLUS Oral Doxycycline 200mg q12h Day 1 then 100mg q12h ²		DURATION: 7 days ²

SEPTIC SHOCK - EMPIRICAL ANTIBIOTIC

Non-diabetics	IV Piperacillin-Tazobactam 4.5g q8h	Need to be revised/deescalated after 48h to 72h based on culture(s) result.
Diabetics	IV Ceftazidime 2g q6-8h	
Neutropenic	IV Piperacillin-Tazobactam 4.5g q6h	

CHAPTER 3

HOSPITAL-ACQUIRED INFECTIONS

Hospital-acquired infection (HAI) is also known as healthcare-associated infections or nosocomial infections. They are infections that the patients acquired while or soon after receiving health care, which are not apparently present or might be incubating at the time of admission. These infections are usually acquired after hospitalization and manifest 48 hours after admission to the hospital. It is important to differentiate it from community acquired infection because the types of pathogens and the resistant pattern of those pathogens may significantly differ with those responsible for similar infections acquired in the community.

The pathogens commonly responsible for most HAI are;

GROUP	MICROORGANISMS	TYPES OF RESISTANT PHENOTYPES
Enterobacterales	<i>Klebsiella pneumonia</i> <i>Escherichia coli</i> <i>Serratia marcescenes</i> <i>Proteus mirabilis</i> <i>Enterobacter sp.</i>	Amp.C ESBL KPC OXA MBL
Nonfermenting gram-negative bacilli	<i>Acinetobacter baumannii</i> <i>Pseudomonas aerogenosa</i> <i>Stenotrophomonas maltophilia</i>	MDRAB, XDR AB MDR PAE, CR PAE Intrinsically resistant to carbapenem
GPC in clusters	<i>Staphylococcus aureus</i> <i>S. epidermidis</i>	MRSA, VISA, VRSA MRCONS
GPC in chains	<i>Enterococcus sp.</i> <i>Streptococcus sp.</i>	Penicillin resistant Enterococcus, VRE.

Hospital acquired infections (HAI) include;

- Central line-associated bloodstream infections (CLABSI, CRI or CRBSI),
- Catheter-associated urinary tract infections (CAUTI),
- Surgical site infections (SSI),
- Hospital-acquired Pneumonia (HAP),
- Ventilator-associated Pneumonia (VAP),
- *Clostridium difficile* infections (CDI).

Risk factors for HAI are;

- Prolonged hospital stay
- Immunosuppressive therapy
- Severe disease with multiple comorbidities
- Ventilator use
- The presence of invasive devices

CHAPTER 4

GUIDELINE FOR THE EMPIRICAL ANTIBIOTIC THERAPY FOR HOSPITAL-ACQUIRED INFECTION

DIAGNOSIS		FIRST CHOICE/ PREFERRED CHOICE	ALTERNATIVE	DURATION (DAYS)
HAP/VAP Early onset Definition: 2-4 days of admission/intubation.		IV Amoxicillin-Clavulanate 1.2g q8h ^{1,2,5}	IV Ceftriaxone 2g q24h	7
	HAP/VAP Late onset Definition: 5 days or more of admission/intubation.	IV Piperacillin-Tazobactam 4.5g q8h ^{1,2,5} PLUS/MINUS IV Vancomycin 15-20mg/kg q8-12h	IV Cefepime 2g q8h PLUS /- IV Vancomycin 15-20mg/kg q8-12h	7
UTI/ CAUTI	Cystitis	Oral Nitrofurantoin 50-100mg q6h ²	Oral Cephalexin 500mg q12h	5 ^{2,4}
	Pyelonephritis	IV Amoxicillin -Clavulanate 1.2g q6h	IV Cefuroxime 750mg q8h	7 ²
	Complicated	IV Amoxicillin -Clavulanate 1.2g q6h PLUS/MINUS Aminoglycoside	IV Ceftriaxone 1-2g q24h PLUS /- Aminoglycoside	10-14 ^{2,4}
	Complicated and AKI	IV Piperacillin/Tazobactam ⁴ (refer dose adjustment in Chapter 11)		
CRI/ CRBSI/ CLABSI²	Central catheter	IV Cloxacillin 2g q4-6h		7 to 10 Suggest for catheter removal
	Hemodialy- sis catheter	IV Cloxacillin 2g q4-6h PLUS IV Ceftazidime 2g q24h		10 to 14 Duration depends on organism isolated from blood culture.
	*If risk of MRSA	IV Vancomycin 15-20mg/kg PLUS IV Ceftazidime 2g q24h		

DIAGNOSIS	FIRST CHOICE/ PREFERRED CHOICE	ALTERNATIVE	DURATION (DAYS)
Joint Prosthesis Infections^{1,2} 1. Stop antibiotics until specimen taken for cultures. 2. Empirical treatment is not recommended. 3. Antibiotic therapy is based on culture and sensitivity. 4. Treatment strategy and duration of antibiotic depends on surgical options. Surgical Options: 1.No surgery 2.Joint removal or fusion 3.DAIR 4.Revision; (i) 1-stage revision (ii) 2-stage revision	MSSA IV Cloxacillin 2g q4-6h PLUS Oral Rifampicin 300-450mg q12h Followed by an oral combination based on susceptibility. Rifampicin should be included if the implant is in situ.		14-42 ^{1,2}
	MRSA IV Vancomycin 15-20mg/kg q8-12h PLUS Oral Rifampicin 300-450mg q12h Followed by an oral combination based on susceptibility. Rifampicin should be included if the implant is in situ.		14-42 ^{1,2}

CHAPTER 5

Staphylococcus aureus INFECTIONS

Staphylococcus aureus (Staph) is a bacterium commonly found on the skin and in the nose of about 30% of individuals. It is the most pathogenic of *Staphylococcus* species. *S.aureus* causes the following conditions;

- (1) Superficial skin lesions (boils, styes) and localized abscesses in other sites.
- (2) Deep-seated infections, such as osteomyelitis and endocarditis and more serious skin infections (furunculosis).
- (3) *S.aureus* is a major cause of hospital acquired (nosocomial) infection of surgical wounds - and, with *S. epidermidis*, causes infections associated with indwelling medical devices.
- (4) *S.aureus* causes food poisoning by releasing enterotoxins into food.
- (5) *S.aureus* causes toxic shock syndrome by release of superantigens into the blood stream.

MSSA

Sensitive strains of *S.aureus* (termed Methicillin-sensitive *S.aureus* - MSSA) are common cause of community and hospital-acquired bloodstream and invasive infections. Complicated MSSA infection requires infectious disease specialist consultation:

- a. Endocarditis
- b. Involving prosthetic devices
- c. Infection involving 2 or more organs/systems (disseminated infection)
- d. Positive blood culture > 72 hours after initiating effective antibiotic
- e. Fever > 72 hours after initiating effective antibiotic

MRSA

Staphylococcus aureus may show resistance to commonly used beta-lactam antibiotics. Alteration of penicillin-binding protein PBP2a leads to resistance to all β -lactam antibiotics (penicillins, cephalosporins, beta-lactam/beta lactamase inhibitor combination and carbapenems). This is known as **Methicillin-resistant *S.aureus* (MRSA)**.

The infection rate of MRSA at HPUSM was relatively low, at 10.0 per 1,000 hospital admissions and showed a decreasing trend in recent years. (al-Talib et al, 2010). Risk factors for MRSA include indwelling catheter, ICU admission, prolonged hospitalization, multiple comorbidities and severe underlying disease.

Vancomycin is indicated for empirical therapy if MRSA risk is high, or the condition is severe. Infectious disease consultation is recommended. If antibiotic susceptibility testing showed MRSA, vancomycin is the drug of choice for the treatment. Meticulous hand hygiene and strict contact precautions are essential to prevent their spread.

NOTE: However, vancomycin is inferior in terms of efficacy against MSSA, when compared to anti-staphylococcal penicillins such as cloxacillin, hence should not be used for MSSA.

CHAPTER 6

MULTIDRUG-RESISTANT (MDR) AND EXTENSIVELY DRUG-RESISTANT (XDR) GRAM-NEGATIVE INFECTIONS

MDR organism is defined as an organism which is non-susceptible to at least one (1) antibiotic from at least three (3) different classes of antibiotic.

XDR organism is defined as an organism which is non-susceptible to at least one (1) antibiotic in all but two (2) or fewer classes of antibiotic.

MDR and XDR organisms are resistant to multiple classes of antibiotics resulting in limited choice of antibiotics for the treatment of infection.

Commonly isolated MDR/XDR Gram-negative organisms:

- MDR/XDR *Pseudomonas aeruginosa*
- MDR/XDR *Acinetobacter baumannii*
- MDR/XDR *Enterobacterales*

These organisms may cause hospital-acquired infections in high dependency/intensive care settings.

Meticulous hand hygiene and strict contact precautions are essential to prevent their spread.

CHAPTER 7

GUIDELINE FOR THE TARGETED ANTIBIOTIC THERAPY FOR HA-XDRO

DIAGNOSIS	RISK STRATIFICATION	RECOMMENDED ANTIMICROBIAL(S) THERAPY	DURATION (DAYS)
<i>S.maltophilia</i>	First line	Oral/ IV Trimethoprim-sulfamethoxazole 15mg/kg/day (Trimethoprim component) in 3 to 4 divided doses	7
	Bactrim Contraindicated ^{1,6,7}	IV Ceftazidime 2g q8h ⁸	
AmpC β-lactamase-Producing Enterobacterales (<i>Enterobacter cloacae</i> complex, <i>Klebsiella aerogenes</i> , and <i>Citrobacter freundii</i> also at significant risk of inducible AmpC production)	Mild to moderate infection	UTI⁷ ▪ Uncomplicated Oral Nitrofurantoin 50-100mg q6h OR Oral Trimethoprim-sulfamethoxazole 15mg/kg/day (Trimethoprim component) in 3 to 4 divided doses OR Single dose Gentamicin ▪ Complicated Oral Trimethoprim-sulfamethoxazole 15mg/kg/day (Trimethoprim component) in 3 to 4 divided doses OR Once daily Gentamicin OR Ceftriaxone 1-2g q24h (mild) ▪ Complicated - moderately severe Cefepime 2g q8h(moderate)	7
	Severe infection: -Sepsis > Shock -Delirium -Bacteremia	Normal Renal Function	7
		Impaired Renal Function	
Extended-spectrum B-lactamase-producing Enterobacterales (ESBL-E)⁶	Mild/Uncomplicated UTI	Oral Trimethoprim- sulfamethoxazole 15mg/kg/day (Trimethoprim component) in 3 to 4 divided doses OR Oral Nitrofurantoin 100mg q6h	7

DIAGNOSIS	RISK STRATIFICATION	RECOMMENDED ANTIMICROBIAL(S) THERAPY	DURATION (DAYS)
Continue	Non-severe, outside UTI	IV Amoxycillin-clavulanic acid 1.2g q8h ¹⁶	7
	Severe infection⁶ (SIRS+)	Ertapenem 1g q24h Hypoalbuminemia: #IV Ertapenem 1g q12h	7 to 10
	Critically ill (Intubated, ICU)⁶	# IV Meropenem or Imipenem	7 to 10
	<i>If cefepime OR tazosin was initiated as empiric therapy for cystitis caused by an organism identified as an ESBL-E and clinical improvement occurs, no change or extension of antibiotic therapy is necessary.</i>		

Requires ID physician approval & authorisation.

CHAPTER 8

CARBAPENEM-RESISTANT GRAM-NEGATIVE INFECTIONS

CRE can be defined as members of the Enterobacterales order resistant to at least one carbapenem antibiotic (i.e., ertapenem, meropenem, imipenem, doripenem) or producing a carbapenemase enzyme. Resistance to at least one carbapenem other than imipenem is required for bacteria intrinsically less susceptible to imipenem (e.g., *Proteus*, *Morganella spp.*, *Providencia spp.*).

This CRE organisms belong to a heterogenous group of pathogens encompassing multiple mechanisms of resistance. They can be broadly divided into those that are not carbapenemase-producing and those that are carbapenemase-producing. The CRE which are not carbapenemase-producing may be the result of amplification of non-carbapenemase B-lactamase genes (e.g., ESBL genes) with concurrent outer membrane porin disruption. The Carbapenemase-producing isolates account for 35%-83% of CRE cases in the United States, with higher percentages observed when restricting the definition of CRE to require resistance to meropenem or imipenem.

The most common carbapenemases in Malaysia are New Delhi metallo- β -lactamases (NDMs), whereas in the US, the most common carbapenemases are *K. pneumoniae* carbapenemases (KPCs), which are not limited to *K. pneumoniae* isolates. Other carbapenemases include Verona integron-encoded metallo- β -lactamases (VIMs), imipenem-hydrolyzing metallo- β -lactamases (IMPs), and oxacillinases (e.g., OXA-48-like)²¹⁸⁻²²⁰. NDM, VIM, and IMP carbapenemases are collectively referred to as metallo- β -lactamases (MBLs).

CARBAPENEM RESISTANT GRAM-NEGATIVE BACTERIA

- *Stenotrophomonas maltophilia*
- Carbapenem resistant *Acinetobacter baumannii* (CRAB)
- Carbapenem resistant Enterobacterales (CRE):
 - *Klebsiella pneumoniae*
 - *E.coli*
 - *Enterobacter sp.*
 - *Proteus mirabilis*
- Carbapenem resistant *Pseudomonas aeruginosa*

Initial approach to patients with suspected infection with extensively drug resistant organisms (XDRO):

The Three (3) Steps;

- 1. Determination of the status of the organism(s) isolated:** Infection vs. Colonization vs. Contamination.
- 2. Assess the patient's comorbidity:** Is the patient going to significantly recover from the treatment of this XDRO infection?
- 3. Evaluate clinical severity:** Stable or mild-moderate vs. unstable or severe to critically ill conditions.

STEP 1: Infection vs. Colonization vs. Contamination

INFECTION = Clinical, biochemical, and/or radiological evidence of infection;

- Requires treatment.
- Requires decolonization.
- Requires contact precautions.

COLONIZATION = True culture results but not causing the infection;

- Does not require treatment.
- Require decolonization.
- Requires contact precautions.

CONTAMINATION = The pathogen cultured does not originate from the patient;

- Does not require treatment.
- Does not decolonization.
- Requires contact precautions.

STEP 2: Assessment of severity or risk of rapid deterioration. Determine whether the patient is clinically stable or unstable.

Important parameters to consider in the evaluation of the patients in addition to the standard history taking, examination, and investigations, include:

- Presence of bacteremia
- Septic shock
- Respiratory failure
- Inotropic support
- Requires ICU admission
- Increasing ventilator settings

STEP 3: Assessment of the patient's potential to recover from the treatment of this infection

The impact of the underlying illness can be so severe that a recurrent infection is inevitable, and the treatment of one infection will lead to another infection with a more resistant organism or phenotypes. Therefore, the treatment of the patients with XDRO infection is generally not appropriate in the following situations:

- The patient who are terminally ill from a progressive, advance disease.
- The patient is not expected to recover with a reasonable quality of life upon the treatment of the infection.
- There is a no plan to remove the source of infection (i.e., no source control), making definitive cure impossible.

In these situations, patients should be offered symptomatic and supportive care to ensure comfort and minimize symptoms. The discussion with the family members are also strongly recommended for getting the advance directive in the event of further deterioration.

CHAPTER 9

GUIDELINE FOR THE TARGETED ANTIBIOTIC THERAPY FOR CARBAPENEM-RESISTANT GNB

DIAGNOSIS	RISK STRATIFICATION	RECOMMENDED ANTIMICROBIAL(S) THERAPY	DURATION (DAYS)
Carbapenem-Resistant <i>Acinetobacter baumannii</i> (CRAB)⁹	Mild^{7,9}	Susceptible IV Ampicillin-sulbactam 3g q4h Non-susceptible ⁹ IV Ampicillin-sulbactam 9g q8h	7 to 10
	Moderate-severe	IV Ampicillin-sulbactam 9g q8h over 3h infusion PLUS/MINUS #IV Meropenem 2g q8h over 3h infusion	
Carbapenem-Resistant <i>Enterobacterales</i> (CRE)⁷	Uncomplicated cystitis^{6,7}	Oral Nitrofurantoin 100mg q6h OR Single-dose of an aminoglycoside	7 to 10
	Complicated cystitis	Oral Trimethoprim-sulfamethoxazole 15mg/kg/day (Trimethoprim component) in 3 to 4 divided doses OR Oral Ciprofloxacin 500-750mg q12h	
	Outside Urinary Tract		
	Mild to moderate	Oral Trimethoprim-sulfamethoxazole 15mg/kg/day (Trimethoprim component) in 3 to 4 divided doses OR Oral Ciprofloxacin 500mg q8h	
	Severe	#IV Meropenem 2g q8h infusion PLUS IV Gentamicin 7mg/kg q24h. OR #IV Polymyxin B 12,500IU/kg q12h OR #IV Tigecycline 100mg q12h	
MDR/XDR or DTR <i>P.aeruginosa</i>	Cystitis	Single dose IV Gentamicin 5mg/kg	Stat
	Mild-moderate	IV Gentamicin 5-7mg/kg daily OR IV Ceftolozane-Tazobactam 3g q8h ¹⁰ AND/OR Any antibiotic with in-vitro activity (Sensitive)	7
	Severe	IV Gentamicin 5-7mg/kg daily PLUS IV Ceftolozane-Tazobactam 3g q8h ¹⁰ OR Any antibiotic with in-vitro activity (sensitive)	7 to 10

Requires ID physician approval & authorisation.

CHAPTER 10

DOSAGE REGIMEN OF SULBACTAM FOR *Acinetobacter* MDR

SULBACTAM RATIO

IV Unasyn 1.5g/ Vial	1g Ampicillin + 0.5g Sulbactam (Ratio, 1:0.5)
IV Sulperazone 1g/ Vial	Cefoperazone 500mg + Sulbactam 500mg (Ratio, 1:1)

DOSE

Antimicrobial	Adjustment Dose for Renal Failure (CrCl ml/min)			CVVH/CVVHD/ CVVHDF
	>50	20-50	<20/HD	
Target Sulbactam per day	9g	6g	4g	9g
Unasyn [®]	9g q8h	6g q8h	6g q12h	9g q6h
Sulperazone [®]	4g q6h	4g q8h	4g q12h	4g q6h

ADMINISTRATION

Dilution: 100 cc
 Diluent: Water for Injection/ Normal Saline 0.9%/ Dextrose 5%
 Infusion: Over 4 hours

SPECIAL PRECAUTIONS

Unasyn [®]	High doses may cause hypernatraemia • 1 vial of 1.5g IV Unasyn, contains 5mmol of Na+ • E.g., IV Unasyn 9g q8h provides 90mmol Na+ • May dilute with Sodium Chloride 0.18% & Dextrose 4.25% (QSD1), Dextrose 5%
Sulperazone [®]	High doses may cause Vitamin K deficiency. • Cefoperazone contains a N-methylthiotetrazole (NMTT) side chain which can inhibit vitamin K metabolism causing hypoprothrombinemia and PT prolongation • Monitoring parameters: FBC, INR, prothrombin time, hematopoietic symptoms • May administer exogenous IV Vitamin K 10mg OD while on IV Sulperazone

CHAPTER 11

ANTIMICROBIAL RENAL DOSING

ANTIMICROBIAL / DOSE FOR NORMAL RENAL FUNCTION	ADJUSTMENT DOSE FOR RENAL FAILURE (CRCL ML/MIN)			DIALYSIS DOSE
	> 50	10 - 50	< 10	
ACYCLOVIR (IV) 5 - 12.5mg/kg q8h	NA	5 - 12.5mg/kg q12-24h	5 - 6.25mg/kg q24h	CAPD/HD: 2.5 - 6.25mg/kg q24h
AMOXICILLIN (PO) 250mg - 1g q8h	NA	250mg - 1g q8-12h	250mg - 1g q24h	CAPD: 150mg/L HD: 250 - 500mg q24h
AMOXICILLIN/CLAVULANATE 1.2g q8h * in serious infection can increase to q6h	NA	1.2g q12h		CAPD: No data HD: 1.2g q12h
AMPHOTERICIN B 0.25 - 1mg/kg q24h * Amphotericin B lipid complex 5mg/kg q24h - no adjustment needed		NA	0.5 - 0.7mg/kg q24-48h	HD: 0.5 - 1mg/kg q24h (after HD on HD day)
AMPICILLIN 1 - 2g q4-6h	NA	30 - 50ml/min: 1 - 2g q6-8h 10 - 30ml/min: 1 - 2g q8-12h	1 - 2g q12h	CAPD: 125mg/L HD: 1 - 2g q12h
AMPICILLIN/SULBACTAM 1.5 - 3g q6h	NA	1.5 - 3g q8-12h	1.5 - 3g q24h	CAPD: LD: 1.5g/bag MD: 200mg/bag HD: 3g q24h (post HD on HD day)
CEFEPIME 1 - 2g q8-12h	1 - 2g q12h	500mg - 2g q24h	250mg - 1g q24h	IPD: 1g q24h CAPD: LD: 500mg/L MD: 125mg/L HD: 1g q24h + 1g post HD
CEFOPERAZONE 1 - 2g q12h		NA		CAPD: LD: 500mg/L MD: 62.5 - 125mg/L
CEFTAZIDIME 1 - 2g q8h	1-2g q8-12h	1 - 2g q12-24h	1 - 2g q24-48h	IPD: 1 - 1.5g q24h (long dwell), 20mg/kg (short dwell) CAPD: LD: 500mg/L MD: 125mg/L HD: 2g q24-48h
CEFUROXIME 750mg - 1.5g q8h * in serious infection can increase to 1.5g q6h	NA	30-50ml/min: No adjustment 10-30ml/min: 750mg - 1.5g q12h	750mg - 1.5g q24h	CAPD/HD: 750mg - 1.5g q24h
CIPROFLOXACIN 400mg q12h	NA	30 - 50ml/min: No adjustment 10 - 30ml/min: 200 - 400mg q24h	400mg q24h or 200mg q12h	CAPD: 50mg/L HD: 400mg q24h
CLARITHROMYCIN (PO) 500mg q12h	NA	250 - 500mg q12h	250 - 375mg q12h	CAPD/HD: 500mg q24h
COLISTIN LD: IBW > 75kg: 9MU IBW 61-74kg: 8MU IBW 51-60kg: 7MU IBW < 50kg: 6MU MD: 4.5MU q12h	4.5MU q12h	30 - 50ml/min: 3MU q12h 10 - 30ml/min: 2.5MU q12h	2MU q12h	HD: 2MU q12h on non HD days then add 1.5MU post HD CAPD: LD 9MU; MD 4.5-6MU q24h
ERTAPENEM 1g q24h	NA	31 - 50ml/min: No adjustment <30ml/min: 500mg q24h	500mg q24h	CAPD/HD: 500mg q24h
ETHAMBUTOL (PO) 15 - 25mg/kg q24h	NA	30-50ml/min: No adjustment 15-25ml/min: 15-25mg/kg q48h	15mg/kg q48h	CAPD/HD: 15mg/kg q48h

FLUCONAZOLE (PO/IV) 100 - 400mg q24h	NA	50 - 200mg q24h	IPD: 200mg q24-48h CAPD: LD 200mg; MD 50 - 100mg q24h
GANCICLOVIR Induction: 5mg/kg q12h MD: 5mg/kg q24h or 6mg/kg 5x/week	50-69ml/min: 2.5mg/kg q12h	25 - 49ml/min: 2.5mg/kg q24h 10 - 24ml/min: 1.25mg/kg q24h	1.25mg 3x/week CAPD/HD: 1.25mg/kg 3x/week
IMIPENEM/CILASTIN 500mg - 1g q6h	NA	31-50ml/min: 250-500mg q8h 10-30ml/min: 250-500mg q12h	125 - 250mg q12h IPD: 500mg in alternate exchange CAPD: LD 250mg/L; MD 50mg/L HD: Dose after dialysis
ITRACONAZOLE (PO) 100 - 200mg q12h	NA	50 - 100mg q12h	CAPD/HD: 100mg q12-24h
LAMIVUDINE (PO) 300mg q24h	NA	30-49ml/min: 150mg q24h 15-29ml/min: 150mg 1st dose, then 100 - 150mg q24h	150mg 1st dose, then 50 - 75mg q24h CAPD/HD: 50mg 1st dose the 25 - 75mg q24h
LEVOFLOXACIN (IV/PO) 750mg q24h	NA	31 - 50ml/min: 750mg q48h 10 - 30ml/min: 750mg then 500mg q48h	750mg then 500mg q48h CAPD/HD: 750mg q24h then 500mg q48h
MEROPENEM 500mg - 2g q8h	NA	500mg - 2g q12h	500mg - 2g q24h IPD: 1g q24h CAPD: MD 125mg/L HD: 500mg q24h
METRONIDAZOLE 250 - 500mg q8-12h	NA	250 - 500mg q12h	HD: 250 - 500mg q12h
NITROFURANTOIN 50 - 100mg q6h	51-60ml/min: Avoid use	Avoid use	
PENTAMIDINE (IV/IM) 4mg/kg q24h	NA	4mg/kg q24-36h	CAPD: 4mg/kg q24-36h HD: 4mg/kg q48h
PIPERACILLIN/TAZOBACTAM 4.5g q6h	NA	20-40ml/min: 2.25-3.375g q6h 10-20ml/min: 2.25g-3.375g q8h	2.25g q6-8h CAPD: LD 4.5g/bag; MD 1.125 g/bag HD: 2.25g q6h
PYRAZINAMIDE (PO) 25mg/kg q24h (max 2.5g OD)	NA	21-50 ml/min: No adjustment 10-20ml/min: 25mg/kg q48h	25mg/kg q48h (3x/week) CAPD: 25mg/kg q24h HD: 25mg/kg q48h
STREPTOMYCIN (IM) 15mg/kg q24h or 25 - 30mg/kg 2x/week or 25 - 30mg/kg 3x/week	15mg/kg q24h	15mg/kg q24-72h	15mg/kg q72-96h HD: 15mg/kg q72-96h post HD
TRIMETHOPRIM (TMP)/ SULPHAMETHOXAZOLE (SMZ) 5 - 20mg/kg/day of TMP base q6-12h	NA	31-50ml/min: 5mg/kg/day q8-12h 10-30ml/min: 5-10mg/kg/day q12h	5-10mg/kg/day q24h ND
VORICONAZOLE IV: 6mg/kg q12h Day 1 then 4mg/kg q12h onward PO: >40kg: 400mg q12h Day 1 then 200mg q12h <40kg: 200mg q12h Day 1 then 100mg q12h	NA	IV: Avoid use (prefer PO formulation) PO: No adjustment	IPD: 2.5mg/kg q24h CAPD/HD: No data, prefer oral route to avoid accumulation of IV cyclodextrin vehicle

HD (Hemodialysis); IPD (Intermittent Peritoneal Dialysis); LD (Loading Dose); MD (Maintenance Dose); CAPD (Continuous Ambulatory Peritoneal Dialysis); ND (No Data); NA (No Adjustment)

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